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Verifying the Prohibition on the Placement of Nuclear Weapons in Outer Space

Legal, Technical, and Policy Pathways

Introduction

The growing dependence of military, civilian, and commercial systems on space infrastructure, combined with renewed concerns about the possible deployment of nuclear weapons in orbit, has revived interest in strengthening the Outer Space Treaty (OST). While Article IV of the OST clearly prohibits the placement, installation, and stationing of weapons of mass destruction in orbit, it contains no verification mechanism. This gap is becoming increasingly concerning because a nuclear anti-satellite (ASAT) weapon would have extremely negative effects on the space environment, degrading entire orbital regimes for potentially long periods of time, indiscriminately damaging satellites, and creating dangerous escalation pressures on Earth. Recognizing these challenges, the Council on Strategic Risks (CSR), the Geneva Centre for Security Policy (GCSP), and the Secure World Foundation (SWF) convened a Track 1.5 dialogue in Geneva in May 2026 to examine legal, technical, and policy pathways for verifying compliance with the OST.

Differentiating from nuclear weapons on ballistic missiles

During the initial discussions of this issue, some colleagues pointed out the need to distinguish a nuclear weapon stationed in space from a nuclear-

tipped intercontinental ballistic missile (ICBM). This discussion then focused on the policy and legal distinctions between the two. While both have significant strategic implications, the illegality of the first under the OST needs to be highlighted. Unlike an ICBM, an orbital nuclear weapon would remain persistently deployed, reducing warning times, ambiguating intent and attribution, creating “use-or-lose” pressures, and potentially offering first-mover advantages before or during a crisis. Participants emphasized that these characteristics make nuclear weapons in space uniquely destabilizing and potentially capable of triggering an escalatory new arms race in outer space.

Technically, a high altitude nuclear detonation would have multiple effects on the space environment, some immediate, others delayed and longer-lasting, depending on its location and yield. The initial blast releases ionizing radiation (mostly as x-rays, as well as gamma radiation and neutrons) that affects spacecraft within line of sight. At the same time, radioactive residuals from the weapon emit electrons. In space, these charged electrons quickly become trapped in the Earth’s magnetic field, amplifying existing radiation belts, where they remain for weeks to years. Repeated exposure to ionizing radiation can permanently disrupt, degrade, or damage electronics onboard satellites passing through these high radiation areas. Thus, both prompt and delayed effects of a nuclear detonation in space pose a severe risk to satellites in orbit.

Differentiating nuclear weapons in orbit from nuclear energy systems

Another distinction raised during participants' initial discussion of verification is the ability to distinguish nuclear weapons in orbit from legitimate nuclear power systems, such as radioisotope thermoelectric generators (RTGs), fission reactors, or developmental nuclear propulsion technologies. Several participants confirmed that a variety of existing and emerging verification capabilities could distinguish between nuclear power sources and nuclear weapons in orbit, provided sufficient consideration is given to resolution, shielding, and standoff distance. It was suggested that companies working on such nuclear power sources be transparent about their activities to demonstrate they are not developing nuclear weapons for space. These companies may be another source of support for a verification capacity that could distinguish peaceful nuclear power from nuclear weapons in orbit.

Agreement on the importance of Article IV of the OST

From a legal perspective, all participants agreed that Article IV of the OST remains the cornerstone of the international regime governing nuclear weapons in space. Additional legal constraints arise from the Limited Test Ban Treaty (also known as the Partial Test Ban Treaty), the UN Charter, broader principles of international law incorporated through Article III of the OST, the Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (Environmental Modification Convention), and the Comprehensive Nuclear Test Ban Treaty. It was widely understood that stationing a nuclear weapon in orbit would violate Article IV of the OST and that any use of such a weapon would certainly implicate multiple legal obligations, particularly in relation to the prohibition of the use or threat of force enshrined in Article 2(4) of the UN Charter. However,

important questions remain unresolved, including whether the mere presence of a nuclear weapon in orbit constitutes an unlawful "threat of force" and how states should interpret concepts such as "weaponization" in practice.

The core challenge is that the OST functions as, in the words of one of the participants, "an arms control treaty without a verification regime." Participants therefore focused on practical methods of strengthening compliance without reopening the treaty itself, which was widely viewed as politically unrealistic. Three broad approaches emerged. The first would entail negotiating a formal verification protocol, although few considered this feasible in the current geopolitical environment. The second would involve developing implementation mechanisms and common interpretations under the existing treaty framework. The third—and possibly most promising—would focus on building customary international law to allow for verification through state practice ("practice of the parties"), transparency measures, declarations, and operational norms that reinforce the prohibition over time.

Verification models

Several verification pathways were discussed. National technical means (NTM), analogous to those used in traditional arms control agreements, could provide independent monitoring capabilities. But NTM could face credibility challenges because only a handful of countries currently possess advanced space situational awareness (SSA) systems. Also, while SSA helps identify objects in orbit, it does not allow verification of payloads. At the other extreme, an international verification organization could conduct inspections and adjudicate compliance disputes, though political acceptance of such an organization remains doubtful. A hybrid approach for sharing SSA data received the greatest support:

combining national capabilities, commercial data, and international coordination through organizations such as the United Nations Office for Outer Space Affairs (UNOOSA) or a future international SSA information coordination mechanism.

The participants also discussed specific technologies. Several colleagues noted that some technical verification options rely on tradeoffs between standoff distance and time, where the closer the inspector satellite can get, the shorter the amount of time is needed to complete the verification. Discussions also revealed that the presented technical options containing tradeoffs between standoff distance and time would best fit cooperative verification scenarios. For example, X-ray radiography requires X-rays to be sent between two satellites and the satellite they are inspecting; all three must remain co-linear for that to be effective, and thus such verification would be most appropriate for a cooperative scenario. Uncooperative satellite inspections, on the other hand, would entail inherent escalation risks; although some participants noted that public announcement, transparency, and multilateral approaches might mitigate these risks. Several experts noted that using a gamma spectrometer would allow one inspector satellite to use passive detection to see a payload and thus could be done in an uncooperative scenario.

From a technical perspective, several participants concluded that verification is challenging but increasingly feasible. Options include pre-launch verification, such as launch-site inspections and challenge inspections, as well as on-orbit verification using proximity operations, neutron detection, gamma spectroscopy, radiographic techniques, and pattern-of-life analysis. However, no single method is likely to suffice. Instead, a multi-modal approach combining SSA data, radiation signatures, inspections, and commercial sensor networks would provide the highest confidence. Advances in commercial space capabilities, AI-enabled data fusion, and globally distributed sensor

networks are making these approaches more accessible than in previous decades.

Additionally, some participants questioned the value of a US government-led verification effort that adversaries would claim was rigged. Thus, sharing verification technology, or at least observational data, with neutral actors could help them independently verify and reach conclusions about the satellite's payload.

The value of verification as deterrence, not just enforcement

Many participants suggested that the greatest value of verification may not be enforcement but deterrence. The ability to detect, localize, characterize, and attribute a prohibited weapon could discourage states from attempting deployment in the first place. Historical examples such as the Vela satellites (12 satellites placed at an altitude of over 100,000 km from Earth in order to monitor compliance with the LTBT) demonstrate how attribution capabilities can reinforce treaty compliance. Verification becomes most effective when paired with clear accountability policies that specify what happens after a violation is detected, including inspections, consultations, public disclosure, and potential responses. The simple expectation of discovery may itself serve as a powerful deterrent.

Understanding and reducing incentives for violating the OST

Some participants emphasized the need to understand the state's motivations for a nuclear ASAT program, as this could help deter its deployment. The key question is not simply whether a state could deploy such a capability, but why it would choose to do so. Possible incentives include the increasing proliferation of very large satellite constellations

that may not be entirely civilian in nature, and that could be quickly degraded or destroyed with this technology, responding to space-based missile defenses, strengthening deterrence, or pursuing warfighting advantages.

Over the last decade, we have seen a shift towards distributed space systems and away from monolithic, exquisite ones. We have also seen increased military reliance on commercial space services. These changes to the world's space architecture introduce new incentives and calculations, particularly amid evolving strategic competition among major powers. Understanding underlying incentives is critical because verification or attribution alone will not eliminate the drivers that could encourage deployment.

The discussion also highlighted the importance of reducing incentives to violate the treaty. Some participants noted that if space systems were more resilient to nuclear effects, the military utility of a space-based nuclear weapon would decline. Radiation hardening, distributed satellite architectures, rapid reconstitution capabilities, placing constellations at higher orbits, and more resilient commercial systems could all reduce the attractiveness of such weapons. (While some participants argued that the costs for pre-launch satellite hardening against at least long-term radiation exposure effects have dropped, others felt that they were prohibitive for competing commercial operators.) This line of discussion focused on deterrence by denial of benefits to the attacker in addition to deterrence through the fear of attribution and accountability.

Confidence-building and transparency as low-hanging fruit

Confidence-building and transparency measures were widely viewed as the most achievable near-term steps. These could include enhanced registration

requirements, expanded information sharing, greater transparency regarding satellite operations, and the development of norms governing proximity operations and on-orbit inspections. Such measures would complement technical verification efforts while helping to build trust among states. Some participants acknowledged that such measures have already been recommended (even by consensus) at the international level, but that what is currently missing is any implementation. Others stressed that smaller and emerging spacefaring nations must be included in these discussions, as they often depend heavily on the protections afforded by the OST despite lacking significant counterspace capabilities of their own. Credibility, transparency, and capacity-building were discussed as essential components of any successful verification architecture.

Conclusion: bolstering the OST is essential

The overall conclusion of the dialogue was that preserving and strengthening the OST's prohibition on nuclear weapons in space is both necessary and achievable, but likely not through treaty renegotiation. A more practical path lies in developing a layered system of deterrence through: education about the threat, SSA, advances in technical options to verify whether satellites are carrying nuclear warheads, commercial participation, and evolving state practice that gradually strengthens compliance and transparency. Over time, these measures could combine to create a credible verification and accountability architecture that reinforces Article IV, reduces incentives to deploy nuclear weapons in orbit, and helps preserve outer space as a stable and secure domain for all nations. ■

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Contributors:

Mallory Stewart

Council on Strategic Risks

Almudena Azcárate Ortega

Secure World Foundation

Simon Cleobury

Geneva Centre for Security Policy

Jenifer Mackby

Geneva Centre for Security Policy

Isobel Porteous

Council on Strategic Risks

Jess Rogers

Council on Strategic Risks

Victoria Samson

Secure World Foundation